DEVELOPERS-AGENTS.md

Canonical maintenance, behavior, and render-architecture guide for developers and automated coding agents that add features or maintain `mdexplore`.

Mission

Maintain a fast, reliable Markdown explorer for Ubuntu/Linux desktop with:

- Left-pane directory tree rooted at a target folder.
- Markdown-only file listing (`*.md`).
- Right-pane rendered preview with math and diagram support.
- `^`, `Refresh`, `PDF`, `Add View`, and `Edit` actions.
- Top-right copy-by-color controls for clipboard file operations.
- A pin button before copy-by-color controls to copy the currently previewed file.

Repository Map

- `mdexplore.py` : large main application module (roughly 15k lines) containing `MarkdownRenderer` , `MdExploreWindow` , preview/PDF orchestration, caching, file watching, search, and the direct Python CLI entrypoint.
- `mdexplore_app/constants.py` : shared constants used by the main app and support modules.
- `mdexplore_app/runtime.py` : runtime/config/GPU-print helper functions.
- `mdexplore_app/pdf.py` : PDF footer stamping, blank-page suppression, and PlantUML stderr helpers.
- `mdexplore_app/icons.py` : icon loading/recoloring helpers used by the tree and tab UI.
- `mdexplore_app/tree.py` : extracted tree/model classes (`ColorizedMarkdownModel` , `MarkdownTreeItemDelegate`).
- `mdexplore_app/tabs.py` : extracted custom tab-bar class (`ViewTabBar`).
- `mdexplore_app/workers.py` : background worker classes for preview render, PlantUML, and PDF export.
- `mdexplore.sh` : launcher (venv lifecycle + dependency install + app run).
- `setup-mdexplore.sh` : full bootstrap helper for local setup (venv, vendored assets, Rust Mermaid renderer).
- `requirements.txt` : Python runtime dependencies.
- `vendor/` : vendored local runtime assets (Mermaid JS, MathJax, PlantUML jar, and related renderer sources/binaries when bootstrapped).
- `test/` : sample Markdown/PDF fixtures used for manual smoke testing.
- `README.md` : user-facing setup and usage documentation.
- `UML.md` : higher-level architecture, class, and activity diagrams.
- `mdexplore.desktop.sample` : user-adaptable `.desktop` launcher template.
- `mdexplor-icon.png` : primary app icon asset (preferred).
- `DESCRIPTION.md` : short repository summary and suggested topic tags.
- `LICENSE` : MIT license text.
- Runtime config file: `~/.mdexplore.cfg` (persisted last effective root).
- Runtime color sidecars: per-directory `.mdexplore-colors.json` files for persisted tree highlight colors where writable.
- Runtime view sidecars: per-directory `.mdexplore-views.json` files for persisted view-tab state (multi-view sessions and custom tab labels).
- Runtime preview-highlight sidecars: per-directory `.mdexplore-highlighting.json` files for persistent text highlights created from the preview pane.

Runtime Assumptions

- Python `3.10+` .
- Linux desktop with GUI support.
- `PySide6` and `QtWebEngine` available via pip dependencies.
- `pypdf` and `reportlab` available for PDF page-number stamping.
- Mermaid should prefer a local bundle when available and only use CDN fallback.
- Rust Mermaid backend is optional and uses `mmdr` when `--mermaid-backend rust` is selected.
- MathJax should prefer a local bundle when available and only use CDN fallback.
- Java runtime is expected for local PlantUML rendering.
- VS Code `code` command may or may not be installed; app should fail gracefully when missing.
- The direct Python entrypoint and the shell launcher do not currently default to the same Mermaid backend:
  - `mdexplore.py` defaults `--mermaid-backend` to `js` .
  - `mdexplore.sh` injects `--mermaid-backend rust` unless explicitly overridden.

Entry Points And Hotspots

- `mdexplore.sh` is the primary launcher path. It handles `.desktop` `%u` arguments, venv bootstrapping, runtime import self-heal, local asset discovery, backend defaulting, and one retry with software-render fallback.
- `mdexplore.py` remains the direct Python entrypoint. `main()` parses `PATH`, `--mermaid-backend`, and `--debug`, resolves the effective root, creates `QApplication`, probes GPU availability, and constructs `MdExploreWindow`.
- `MarkdownRenderer` in `mdexplore.py` owns markdown-it configuration, fenced syntax handling, MathJax/Mermaid source selection, PlantUML integration, and the Rust Mermaid preview-vs-PDF SVG fork.
- `MdExploreWindow` in `mdexplore.py` is the main orchestration boundary and highest-blast-radius edit surface.
- Before changing preview/render/PDF flow, inspect these methods in `mdexplore.py`:
  - `_merge_renderer_pdf_mermaid_cache_seed()`
  - `_inject_mermaid_cache_seed()`
  - `_on_preview_render_finished()`
  - `_prepare_preview_for_pdf_export()`
  - `_restore_preview_mermaid_palette()`
- `_set_preview_html()` in `mdexplore.py` has two preview load paths:
  - direct `setHtml(..., base_url)` for normal-sized documents
  - temp-file `load()` fallback for oversized HTML Changes to base URLs, asset injection, JS boot order, or page-load timing must be tested on both paths.
- `TreeMarkerScanWorker` in `mdexplore_app/workers.py` shares sidecar read contracts with the main window code. Keep `.mdexplore-colors.json`, `.mdexplore-views.json`, and `.mdexplore-highlighting.json` formats compatible across both readers/writers.

Core Behavior You Must Preserve

- Without a path arg, default root is loaded from `~/.mdexplore.cfg` if valid; otherwise home directory is used.
- Both entrypoints support optional root path argument:
  - `mdexplore.sh [PATH]`
  - `mdexplore.py [PATH]`
- Tree shows directories and `.md` files only.
- Selecting a Markdown file updates preview quickly.
- GitHub/Obsidian-style markdown callouts (`> [!TYPE]`) should render as styled callout boxes in preview and PDF output.
- Navigating to a previously cached file must still stat-check mtime/size; if changed, it must re-render instead of showing stale cached HTML.
- PlantUML rendering jobs must run off the UI thread to keep the window responsive.
- PlantUML blocks should not block markdown preview; show placeholders first, then progressively replace each diagram as local render jobs complete.
- PlantUML progressive updates should be applied in-place so the preview scroll location is preserved.
- PlantUML progress should persist across file navigation so returning to a file shows completed diagrams without restarting work.
- Preview scroll position should be remembered per markdown file for the current application run.
- While dragging the preview pane's vertical scrollbar, the preview should show an approximate `current line / total lines` indicator beside the scrollbar handle.
- `Ctrl++`, `Ctrl+-`, and `Ctrl+0` should adjust only the preview pane zoom factor, not the tree pane or overall UI scale.
- Preview zoom changes should briefly show a top-center percentage overlay inside the preview area.
- `^` navigates one directory level up and re-roots the tree.
- User-adjusted splitter width between tree and preview should persist across root changes for the current application run.
- Window title reflects effective root scope (selected directory if selected, otherwise current root).
- Effective root is persisted on close to `~/.mdexplore.cfg`.
- Linux desktop identity should remain `mdexplore` so launcher icon matching works (`QApplication.setDesktopFileName("mdexplore")` + desktop `StartupWMClass=mdexplore`).
- `Edit` opens currently selected file with `code`.
- `Add View` creates another tabbed view of the same current document, starting from the current top-visible line/scroll position.
- View tabs default to top-visible source line numbers, are closeable, and are capped at 8 tabs per document.
- View tabs include a left-side mini bargraph indicating each view's position in the current document.
- View tabs should keep a stable soft-pastel color sequence based on open order.
- View tabs are draggable/reorderable; moving tabs must not change their assigned color.
- Right-clicking a view tab should allow editing a custom tab label up to 48 characters, including spaces.
- If a longer tab label is entered, it should be truncated to the first 48 characters rather than rejected.
- If a custom tab label is cleared back to blank, that tab should resume using the default dynamic line-number label.
- When a tab receives a custom label, the app should capture that tab's current scroll position and top visible source line as the saved label-time beginning.
- Right-clicking a custom-labeled tab should also offer `Return to beginning`, which restores that saved label-time location for the tab.
- Relabeling a custom-labeled tab with different text should reset the saved beginning to the tab's current scroll position/top line at relabel time.
- When assigning a new tab color and wrapping the palette, skip any color already used by open tabs.

- Per-document tabs should persist for the app run: switching to another markdown file and back must restore prior tabs, their order, and the previously selected tab.
- View-tab state should also persist across app restarts via per-directory `.mdexplore-views.json`, keyed by markdown filename.
- For custom-labeled tabs, `.mdexplore-views.json` should also persist the saved label-time beginning location used by `Return to beginning`.
- If custom labels make the tab strip wider than the available space, the tab bar should scroll rather than truncating the tab set.
- Only documents with explicit multi-view state or custom tab labels should be written to `.mdexplore-views.json`; untouched single-view documents should fall back to the default one-tab state without sidecar entries.
- The tab strip should remain hidden when there is only one unlabeled default view.
- If only one view remains and it has a custom label, its tab should stay visible so the custom label and `Return to beginning` action remain available.
- Closing that sole remaining custom-labeled tab should clear the custom label and saved beginning, then fall back to the hidden default single-view state.
- `Refresh` button and `F5` both refresh the current directory tree view (new files appear, deleted files disappear) without requiring app restart.
- `PDF` exports the currently previewed markdown rendering to `<source-basename>.pdf` beside the source file, with centered `N of M` numbering at the bottom of each page.
- PDF export should preflight preview readiness (MathJax/Mermaid/fonts) before snapshotting, and apply print-focused math styling to avoid cramped glyphs.
- `MIN_PRINT_DIAGRAM_FONT_PT` in `mdexplore.py` is the tweakable PDF one-page shrink floor for diagram sizing decisions. Lowering it allows tall diagrams (for example Mermaid activity/flowcharts) to stay on one page more aggressively before the exporter falls back to multi-page spill.
- In PDF mode, headed Mermaid and PlantUML sections should attach the heading to the diagram fence itself for pagination purposes, so Chromium cannot leave the heading on one page and push the diagram to the next.
- If the currently previewed markdown file changes on disk, preview should auto-refresh and report that via status bar message.
- Status bar should show progress during long-running operations (preview loading/rendering, PlantUML completion, PDF export) and should not remain blank during idle gaps.
- Search input uses label `Search and highlight:` and includes an explicit in-field `X` clear action.
- Pressing `Enter` in search should bypass debounce and run search immediately.
- Non-quoted search terms should be case-insensitive; quoted terms should be case-sensitive.
- `CLOSE(...)` should be supported in search queries and require all listed terms to occur within `SEARCH_CLOSE_WORD_GAP` content words.
- Function-style operators should accept both no-space and spaced forms before `(` (for example `OR(...)` and `OR (...)`).
- `AND(...)`, `OR(...)`, and `CLOSE(...)` should accept comma-delimited, space-delimited, or mixed argument lists.
- `AND(...)` and `OR(...)` should be variadic (2+ arguments).
- If search is active and directory scope changes, search should rerun against the new directory scope.
- File highlight colors are assigned from tree context menu and persisted per directory.
- Highlight state persists in `.mdexplore-colors.json` files where writable.
- `Clear All` in the context menu recursively removes highlight metadata after confirmation.
- Copy/Clear scope resolves in this order: selected directory, then most recently selected/expanded directory, then current root.

- Clipboard copy must preserve Nemo/Nautilus compatibility (`text/uri-list` plus `x-special/gnome-copied-files`).
- The pin copy action should copy exactly the currently previewed markdown file using the same clipboard MIME compatibility guarantees.
- Preview context menu should offer persistent `Highlight` , `Highlight Important` , and `Remove Highlight` actions for rendered text selections and existing highlighted blocks.
- Preview text highlights should persist per directory in `.mdexplore-highlighting.json` .
- Persistent preview text highlights should support at least two visual kinds: normal and important. Entries in `.mdexplore-highlighting.json` should carry enough metadata to preserve that distinction across runs.
- Persistent preview text highlights should also surface as left-gutter markers in the preview; clicking a marker should jump to the corresponding highlighted block, and taller markers should reflect highlights that span more lines.
- Important-highlight markers should be visually lighter than normal-highlight markers, and selecting part of an existing block should allow converting that selected subrange between normal and important highlighting.
- Preview context menu should keep standard actions and add `Copy Rendered Text` and `Copy Source Markdown` when there is a text selection.
- `Copy Rendered Text` should copy the selected preview text as plain text.
- `Copy Source Markdown` should map preview selection to source markdown line ranges and copy source text (not rendered plain text). If direct mapping fails, it should use selected-text matching and first/last line fuzzy matching against source markdown lines, then fall back to copying the full source file.
- While search is active, opening a matched markdown file should highlight matched terms in yellow in preview and scroll to the first match.
- While search is active, the preview should show yellow scrollbar-side markers for highlighted hits, and clicking a marker should jump to the nearest underlying hit represented by that marker cluster.
- While search is active, matching markdown files in the tree should show a yellow hit-count pill in the left gutter rather than the older triangle badge.
- Markdown files that currently have more than the default single view should show a small light-gray tab badge beside the markdown icon in the tree.
- Markdown files with persisted preview text highlights should also show a separate marker badge in the tree.
- Tree gutter badge order for markdown files is: search hit-count pill, highlight marker, views badge, markdown file icon.
- The markdown file icon and filename must stay horizontally aligned across all markdown rows by using a fixed-width gutter strip and packing only the active badges inside it.
- Named views with saved `Return to beginning` anchors should also appear as color-matched left-side preview gutter markers positioned by their saved home line number. These markers must render above persistent-highlight markers when their positions overlap.

Formal Behavior Rules Model

This section consolidates the former `RULES.md` content into the canonical agent guide. It is a behavioral specification, not a user guide.

Use it with:

- `README.md` for user-facing setup and usage.
- `UML.md` for subsystem structure.
- this file for maintenance guarantees, behavior rules, and render/debug flow.

1. Rule Language

1.1 Modal operators

- `O(x)` means `x` is mandatory.
- `P(x)` means `x` is permitted.
- `X(x)` means `x` is forbidden.

1.2 Rule form

Each rule is written as:

```
R.<DOMAIN>.<N> :: [guard] => modal(requirement) / effect
```

Examples:

```
R.ROOT.01 :: [cli_path is valid_dir] => O(use cli_path as root)
R.CACHE.01 :: [mtime or size changed] => X(reuse cached_html)
R.MERMAID.06 :: [backend = rust and mode = pdf] => X(use cache["auto"])
```

1.3 Precedence notation

- `A > B` means rule set `A` overrides `B` on conflict.
- `first(a, b, c)` means "first defined/valid value in order".
- `state := expr` means state assignment or resolution.

2. Source and Precedence Model

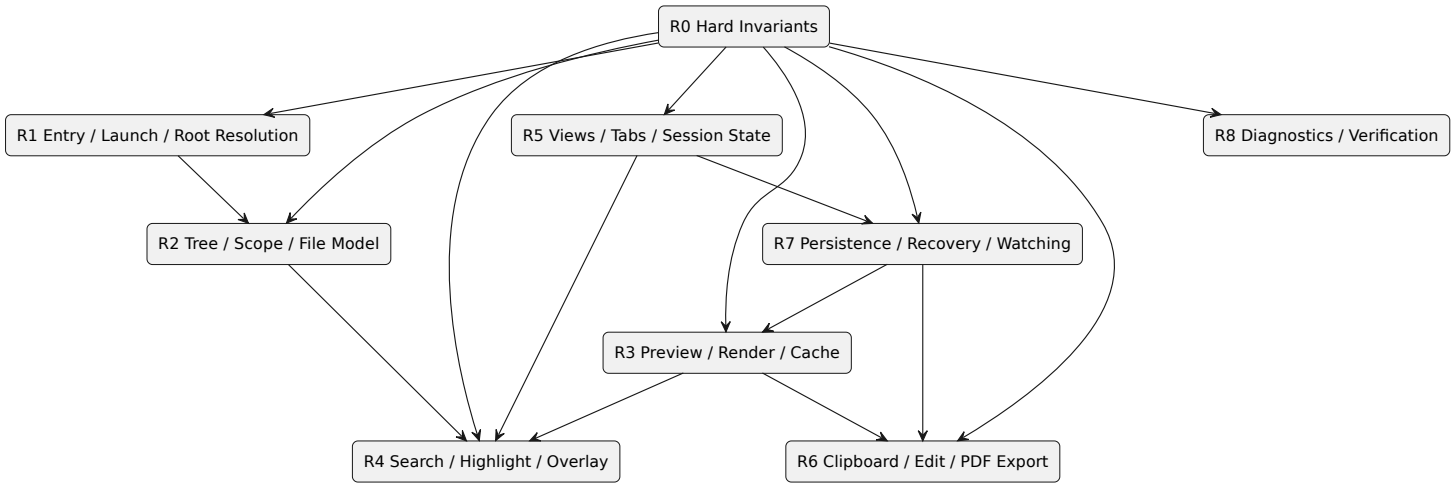
Behavior should be interpreted through the following precedence stack:

```
Current code paths
  > DEVELOPERS-AGENTS.md behavioral guarantees, rules, and render architecture
  > UML.md architectural abstraction
  > README.md user-facing description
```

Within the application itself, runtime rule precedence is:

```
Hard invariants
  > mode/backend-specific safety rules
  > user-visible behavioral guarantees
  > persisted state restoration
  > session-local state
  > defaults
  > fallbacks
```

3. Hierarchy of Rule Domains



Interpretation:

- `R0` defines non-negotiable correctness boundaries.
- `R1-R8` refine behavior by domain.
- No lower tier may violate a higher tier.

4. Global Invariants

These rules are system-wide and dominate every lower rule.

```
R.INV.01 :: [always] => O(show only directories and *.md in the tree)
R.INV.02 :: [always] => O(preserve responsive UI during slow render work)
R.INV.03 :: [always] => O(stat-check cached markdown before reuse)
R.INV.04 :: [always] => O(preserve Linux desktop identity as "mdexplore")
R.INV.05 :: [backend = rust and mode = pdf] => X(use GUI-adjusted SVG as PDF source)
R.INV.06 :: [backend = rust and mode = pdf] => X(select Mermaid cache mode "auto")
R.INV.07 :: [after PDF export] => O(force restore of GUI Mermaid mode "auto")
R.INV.08 :: [PlantUML incomplete] => O(show markdown preview before all diagrams finish)
R.INV.09 :: [search or persisted markers active] => O(treat overlays as post-render
navigation aids, not alternate render outputs)
R.INV.10 :: [file changed on disk] => X(show stale preview as current truth)
```

5. Entry, Launch, and Root Resolution

5.1 Root selection

```
effective_root := first(valid(cli_path), valid(cfg_root), HOME)
```

```
R.ROOT.01 :: [cli_path is valid_dir] => O(use cli_path)
R.ROOT.02 :: [cli_path absent and cfg_root is valid_dir] => O(use cfg_root)
R.ROOT.03 :: [cli_path absent and cfg_root invalid] => O(use HOME)
R.ROOT.04 :: [window closes] => O(persist effective_root to ~/.mdexplore.cfg)
R.ROOT.05 :: [user presses ^] => O(re-root one directory upward)
```

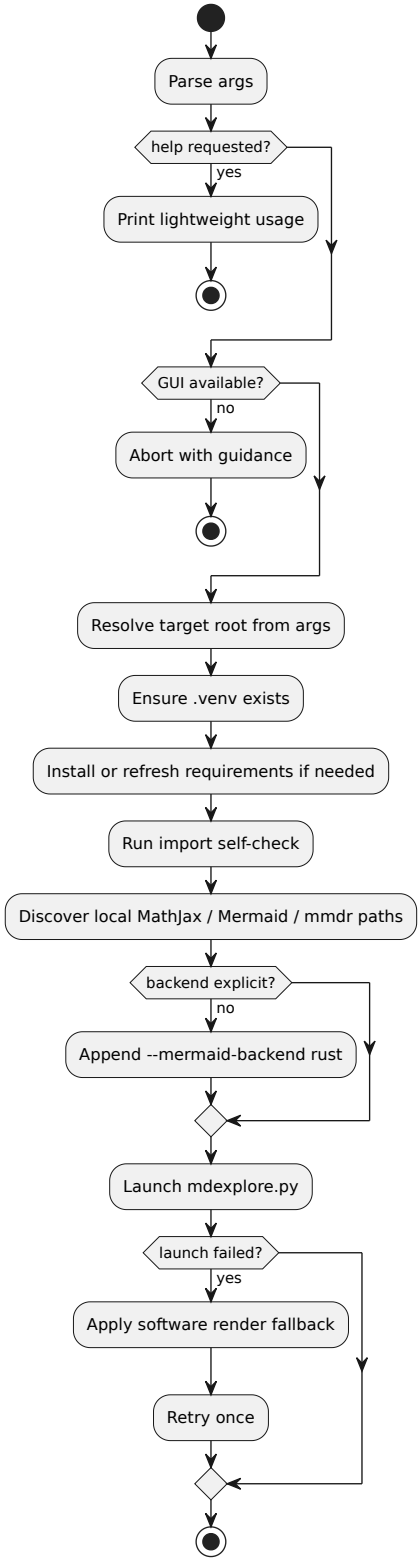
5.2 Entry-point asymmetry

```
launcher_default_backend := rust if no explicit backend
python_main_default_backend := js
```

```
R.ENTRY.01 :: [entry = mdexplore.sh and backend not explicit] => O(append --mermaid-backend
rust)
R.ENTRY.02 :: [entry = mdexplore.py and backend not explicit] => O(default backend to js)
R.ENTRY.03 :: [launcher arg is file:// URI] => O(decode URI)
R.ENTRY.04 :: [launcher arg is a file] => O(resolve parent directory as root)
```

```
R.ENTRY.05 :: [initial GUI launch fails] => O(retry once with software rendering fallback)
R.ENTRY.06 :: [DISPLAY and WAYLAND_DISPLAY absent] => X(start GUI)
```

5.3 Launcher decision model



6. Tree, Scope, and File Model

6.1 Tree constraints

```
R.TREE.01 :: [path is directory] => O(show path in tree)
R.TREE.02 :: [path is file and suffix = .md] => O(show path in tree)
R.TREE.03 :: [path is file and suffix != .md] => X(show path in tree)
R.TREE.04 :: [markdown row rendered] => O(use fixed-width gutter with badge order:
                                          search-hit pill, highlight marker, views badge,
file icon)
R.TREE.05 :: [color highlight exists] => O(paint filename background without breaking row
alignment)
```

6.2 Scope resolution

```
copy_scope := first(selected_directory, last_selected_or_expanded_directory, current_root)
search_scope := highlighted_scope_directory()
```

```
R.SCOPE.01 :: [copy or clear by color] => O(resolve scope by selected dir > last dir >
root)
R.SCOPE.02 :: [search active and directory scope changes] => O(re-run search in new scope)
R.SCOPE.03 :: [tree root changes] => O(preserve session splitter width for the app run)
```

6.3 Scope resolution table

Selected dir	Last dir	Root	Result
yes	any	any	selected dir
no	yes	any	last selected/expanded dir
no	no	yes	current root

7. Preview, Render, and Cache Rules

7.1 Cache law

```
preview_cache_hit(file) :=
  cached(file)
  and cached.mtime_ns = stat(file).mtime_ns
  and cached.size = stat(file).size
```

```
R.CACHE.01 :: [preview_cache_hit(file)] => P(reuse cached HTML)
R.CACHE.02 :: [not preview_cache_hit(file)] => O(re-render file)
R.CACHE.03 :: [cached file revisited] => O(re-stat before trust)
R.CACHE.04 :: [large encoded HTML] => O(use temp-file load path instead of direct setHtml)
```

7.2 Render branch space

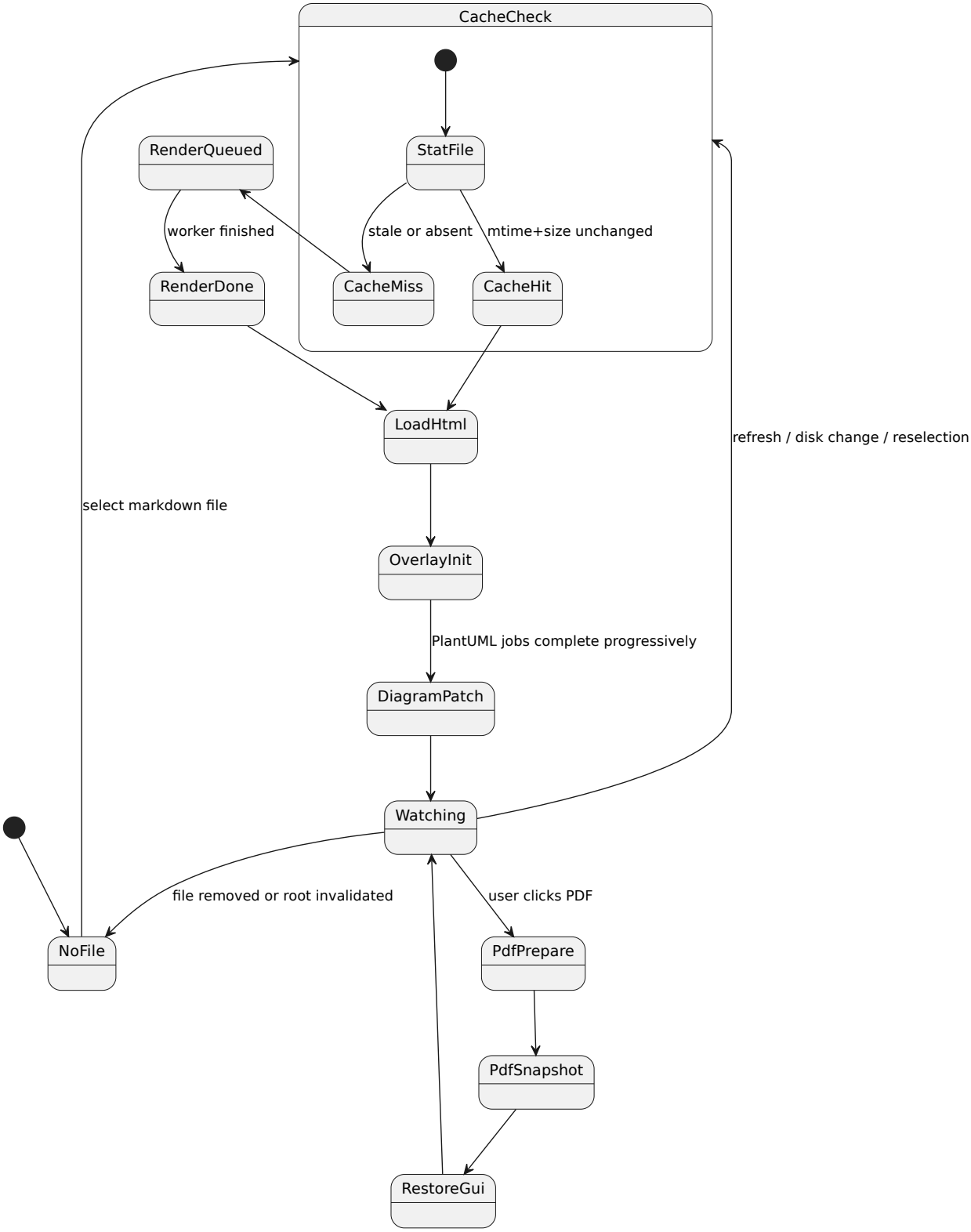
```
mode in {auto, pdf}
backend in {js, rust}
render_branch := (mode, backend)
```

```
R.RENDER.01 :: [backend = js and mode = auto] => O(render Mermaid in page via Mermaid JS)
R.RENDER.02 :: [backend = rust and mode = auto] => O(prefer pre-rendered Rust preview SVG)
R.RENDER.03 :: [backend = rust and preview SVG missing] => P(JS fallback for GUI only)
R.RENDER.04 :: [backend = js and mode = pdf] => O(run PDF clean render and
monochrome/grayscale pass)
R.RENDER.05 :: [backend = rust and mode = pdf] => O(select dedicated Rust PDF SVG)
R.RENDER.06 :: [backend = rust and mode = pdf] => X(reuse GUI SVG or cache["auto"])
R.RENDER.07 :: [PDF finished] => O(restore GUI palette mode to auto)
R.RENDER.08 :: [relative links or images in markdown] => O(preserve base URL behavior)
```

7.3 Math, Mermaid, and PlantUML

```
R.DIAG.01 :: [MathJax available locally] => O(prefer local bundle)
R.DIAG.02 :: [Mermaid available locally] => O(prefer local bundle)
R.DIAG.03 :: [local bundle missing] => P(use CDN fallback)
R.DIAG.04 :: [PlantUML fence detected] => O(render locally via java -jar plantuml.jar)
R.DIAG.05 :: [PlantUML render slow] => O(run off UI thread)
R.DIAG.06 :: [PlantUML unfinished] => O(show placeholders and patch results in place)
R.DIAG.07 :: [PlantUML failed] => O(render inline error with stderr context)
R.DIAG.08 :: [dark preview background] => O(apply high-contrast Mermaid GUI styling)
```

7.4 Preview lifecycle



8. Search, Highlight, and Overlay Rules

8.1 Search semantics

```
case(term) := insensitive if unquoted(term) else sensitive
```

```
R.SEARCH.01 :: [search text empty] => O(clear active match styling immediately)
R.SEARCH.02 :: [typing search text] => O(debounce before running search)
R.SEARCH.03 :: [Enter pressed in search] => O(run search immediately)
R.SEARCH.04 :: [query contains CLOSE(...)] => O(require all terms within
SEARCH_CLOSE_WORD_GAP content words)
R.SEARCH.05 :: [query contains AND(...) or OR(...)] => O(accept comma-delimited, space-
delimited, or mixed arguments)
R.SEARCH.06 :: [query active and file opened from matches] => O(highlight preview hits and
scroll to first match)
R.SEARCH.07 :: [tree row is a match] => O(show hit-count pill in gutter)
```

8.2 Persistent highlights and markers

```
R.HL.01 :: [preview text highlighted] => O(persist to .mdexplore-highlighting.json)
R.HL.02 :: [persistent highlight kind] => O(distinguish normal vs important)
R.HL.03 :: [persistent highlights exist] => O(show preview gutter markers)
R.HL.04 :: [named views have saved homes] => O(show named-view markers above persistent-
highlight markers on overlap)
R.HL.05 :: [file has persistent preview highlights] => O(show dedicated tree marker badge)
```

8.3 Search / marker layering

```
R.OVERLAY.01 :: [search active] => O(show right-side yellow hit markers in preview)
R.OVERLAY.02 :: [persistent highlights active] => O(show left-side highlight markers)
R.OVERLAY.03 :: [named-view home markers active] => O(show above persistent-highlight
markers at overlapping positions)
R.OVERLAY.04 :: [marker clicked] => O(jump to nearest represented target)
```

9. View, Tab, and Session Rules

9.1 Tab constraints

```
R.VIEW.01 :: [Add View invoked] => O(create view at current top-visible line)
R.VIEW.02 :: [per-document view count] => X(exceed 8 tabs)
R.VIEW.03 :: [tab moved] => O(preserve assigned color)
R.VIEW.04 :: [tab label length > 48] => O(truncate to first 48 chars)
R.VIEW.05 :: [custom label cleared] => O(revert to dynamic line-number label)
R.VIEW.06 :: [custom label set or changed] => O(capture current scroll and top line as
label-time home)
R.VIEW.07 :: [custom-labeled tab context menu] => O(expose Return to beginning)
R.VIEW.08 :: [single remaining tab is unlabeled default] => O(hide tab strip)
R.VIEW.09 :: [single remaining tab is custom-labeled] => O(keep tab visible)
R.VIEW.10 :: [closing sole remaining custom-labeled tab] => O(clear custom label and home,
then fall back to hidden default state)
```

9.2 Persistence law

```
persist_view_session(file) := has_multiple_views(file) or has_custom_label(file)
```

```
R.SESSION.01 :: [switch away and back during app run] => O(restore tabs, order, and
selected tab)
R.SESSION.02 :: [persist_view_session(file)] => O(write file state to .mdexplore-
views.json)
R.SESSION.03 :: [not persist_view_session(file)] => X(write default single-view sidecar
entry)
R.SESSION.04 :: [named home anchor exists] => O(persist anchor for Return to beginning)
```

10. Clipboard, Edit, and PDF Export Rules

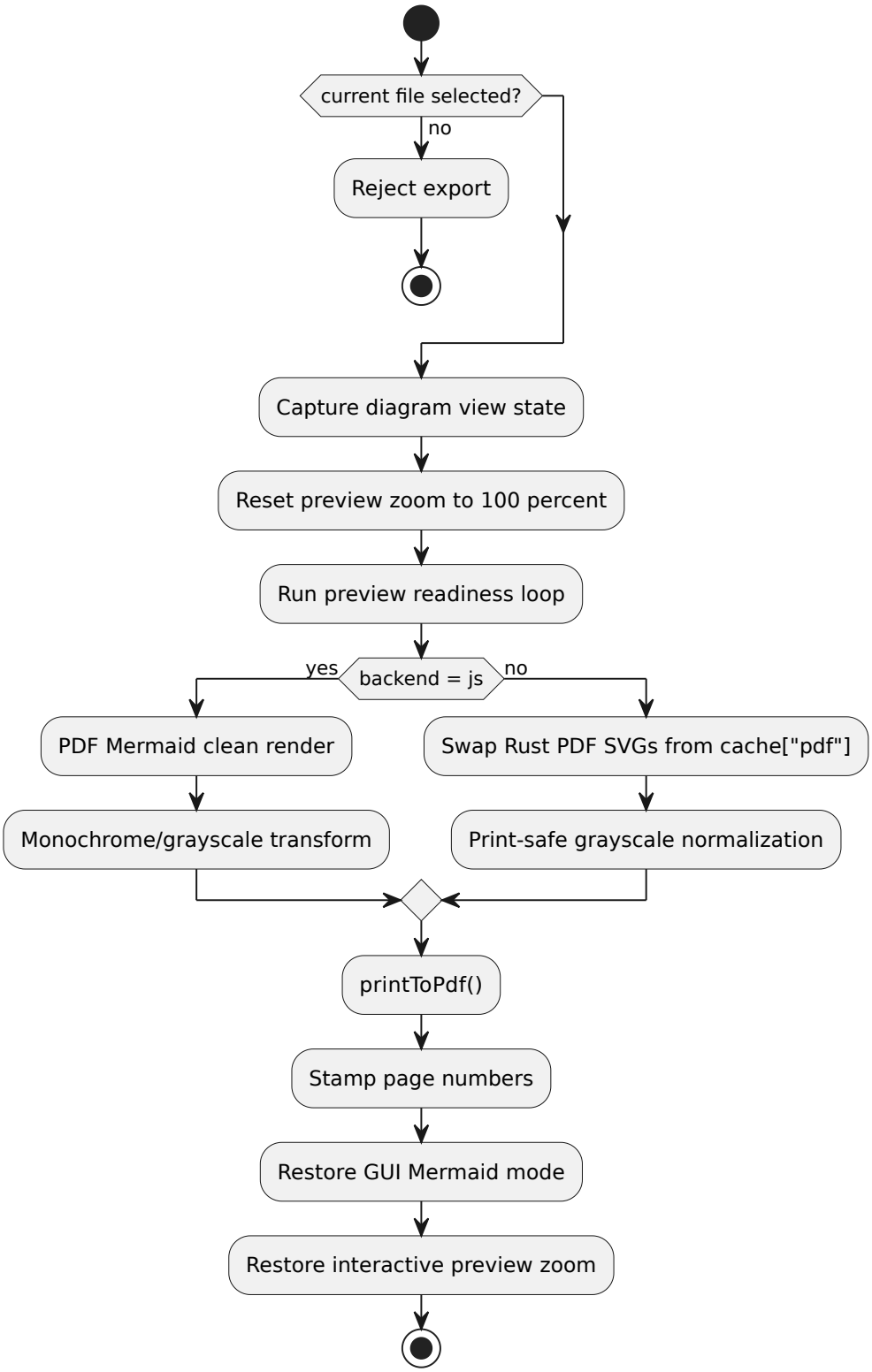
10.1 Clipboard and editor actions

```
R.CLIP.01 :: [copy files by color] => O(write text/uri-list and x-special/gnome-copied-
files MIME forms)
R.CLIP.02 :: [pin action used] => O(copy exactly the currently previewed markdown file)
R.CLIP.03 :: [preview text selection and Copy Rendered Text] => O(copy plain rendered text)
R.CLIP.04 :: [preview text selection and Copy Source Markdown] => O(map selection to source
lines if possible)
R.CLIP.05 :: [direct source mapping fails] => P(fallback to text match, then fuzzy lines,
then full file)
R.CLIP.06 :: [Edit clicked] => O(invoke code on current file)
R.CLIP.07 :: [code missing] => O(fail gracefully with user-visible error)
```

10.2 PDF export law

```
R.PDF.01 :: [current_file absent] => X(start PDF export)
R.PDF.02 :: [PDF export begins] => O(preflight math, Mermaid, fonts, and print CSS
readiness)
R.PDF.03 :: [PDF export begins] => O(capture diagram state and reset preview zoom to 100
percent)
R.PDF.04 :: [PDF snapshot ready] => O(write <source-basename>.pdf beside source file)
R.PDF.05 :: [PDF post-process] => O(stamp centered "N of M" footer numbering)
R.PDF.06 :: [headed Mermaid or PlantUML section] => O(bind heading to diagram fence for
pagination)
R.PDF.07 :: [PDF export ends] => O(restore preview Mermaid mode and interactive zoom)
```

10.3 Export decision chain



11. Persistence, Refresh, and Recovery Rules

```
R.PERSIST.01 :: [tree colors modified] => O(persist per-directory .mdexplore-colors.json
where writable)
R.PERSIST.02 :: [preview text highlights modified] => O(persist per-directory .mdexplore-
highlighting.json)
R.PERSIST.03 :: [current file changes on disk] => O(auto-refresh preview and report it in
the status bar)
R.PERSIST.04 :: [Refresh button or F5] => O(refresh current tree view without restart)
R.PERSIST.05 :: [status bar idle gap] => X(leave status bar blank)
```

12. Diagnostics and Verification Rules

```
R.LOG.01 :: [debug mode off] => O(keep mdexplore.log disabled)
R.LOG.02 :: [debug mode on] => O(write bounded debug log in project root)
R.LOG.03 :: [launcher non-interactive] => O(log to ~/.cache/mdexplore/launcher.log with
retention cap)
R.QA.01 :: [agent changes behavior] => O(run bash -n mdexplore.sh)
R.QA.02 :: [agent changes behavior] => O(run python3 -m py_compile mdexplore.py)
R.QA.03 :: [render behavior changed] => O(re-test GUI and PDF for JS and Rust branches)
```

13. Canonical Decision Tables

13.1 Root resolution

CLI path valid	Config root valid	Result
yes	any	CLI path
no/absent	yes	config root
no/absent	no	HOME

13.2 Mermaid backend default

Entry path	Explicit backend supplied	Effective default
`mdexplore.sh`	yes	explicit value
`mdexplore.sh`	no	`rust`
`mdexplore.py`	yes	explicit value
`mdexplore.py`	no	`js`

13.3 View-session persistence

Multi-view	Custom label	Persist to `.mdexplore-views.json`
no	no	no
yes	no	yes
no	yes	yes
yes	yes	yes

14. Rule Summary

The application can be read as this composite law:

```
mdexplore :=
  markdown_tree_browser
  + cached_but_stat-validated_preview
  + local-first_math_and_diagrams
  + async_PlantUML_progressive_patch
  + mode-separated_Mermaid_cache(auto, pdf)
  + persistent_search/highlight/view_navigation_state
  + Linux-compatible_clipboard_and_editor_actions
  + print-safe_PDF_export_with_restore
```

If one rule must dominate all others, it is this:

```
R.PRIME.01 :: [always] => O(preserve correctness and responsiveness without crossing branch
boundaries)
```

Editing Rules

- Keep code ASCII unless file already requires Unicode.
- Prefer type hints and straightforward, explicit control flow.
- Avoid large framework changes unless requested.
- Do not add heavy dependencies without clear user value.
- Keep startup and preview interactions responsive.
- Preserve cache semantics unless changing performance behavior intentionally.
- Treat `mdexplore_app/` as the extraction boundary for low-risk support code:
  - leaf helpers belong there first,
  - reusable UI support classes that do not own application orchestration may also move there (`tree.py`, `tabs.py`),
  - keep `mdexplore.py` responsible for main-window orchestration and UI state unless a second-stage refactor is explicitly intended,
  - do not duplicate helpers or support classes between `mdexplore.py` and `mdexplore_app/*`.

Rendering Rules

- Markdown parser is `markdown-it-py`.
- Mermaid and MathJax are rendered client-side in web view.
- Mermaid backend is switchable via CLI:
  - `--mermaid-backend js` (default)
  - `--mermaid-backend rust` (renders SVG with `mmdr`)
- Mermaid loading order is local-first, then CDN fallback.
- Mermaid rendering should use a dark-theme-friendly high-contrast palette when preview background is dark.
- Mermaid SVG render results should be cached in-memory (by diagram hash and render mode) for the current app run so revisiting documents avoids rerender.
- Mermaid cache modes are:
  - `auto`: GUI/interactive preview mode.
  - `pdf`: print/export mode.
- PDF Mermaid behavior is backend-specific:
  - JS backend: uses PDF clean render + print-safe monochrome/grayscale transform.
  - Rust backend: uses separate PDF SVGs rendered by `mmdr` with default theming, then applies print-safe grayscale normalization (with gradation and readable text). GUI preview post-processing must not leak into Rust PDF source SVG selection.
- `MDEXPLORE_MERMAID_JS` can be used to force a specific local Mermaid script path.
- `MDEXPLORE_MERMAID_RS_BIN` can be used to force a specific `mmdr` executable path.
- MathJax loading order is local-first, then CDN fallback.
- `MDEXPLORE_MATHJAX_JS` can be used to force a specific local MathJax script path.
- Callout syntax is parsed from blockquotes using `[!TYPE]` markers and rendered as non-collapsible callout containers.
- PlantUML fences are rendered locally through `plantuml.jar` (`java -jar ...`), defaulting to `vendor/plantuml/plantuml.jar` unless `PLANTUML_JAR` is set.
- PlantUML failures should render inline as: `PlantUML render failed with error ...`, including detailed stderr context (line numbers when available).
- Maintain base URL behavior (`setHtml(..., base_url)`) so relative links/images resolve.
- If adding new embedded syntaxes, implement via fenced code handling and document it.
- Debug logging to project-root `mdexplore.log` must remain disabled unless the app is launched with `--debug`.

Mermaid Render/Caching Architecture (Read Before Changes)

This code has multiple render forks. Preserve these boundaries. For practical, symptom-driven troubleshooting, also read section `Render Triage, Architecture, And Debugging` below.

Python-side responsibilities

- `MarkdownRenderer` parses markdown fences and renders Mermaid differently by backend.
- When backend is `rust`:
  - Preview SVG is rendered using Rust preview profile and embedded in HTML.
  - A second PDF-profile SVG (default `mmdr` theming) is rendered per Mermaid hash and stored in `env["mermaid_pdf_svg_by_hash"]`.
- `render_document()` captures this per-document Rust PDF map and exposes it through `take_last_mermaid_pdf_svg_by_hash()`.
- `MdExploreWindow._merge_renderer_pdf_mermaid_cache_seed()` merges that map into runtime Mermaid cache mode `pdf`.
- `_inject_mermaid_cache_seed()` injects the full mode cache (`auto` + `pdf`) into page JS.

In-page JS responsibilities

- `window.__mdexploreMermaidSvgCacheByMode` stores mode-specific SVG caches.
- GUI path:
  - JS backend renders Mermaid through Mermaid JS and stores SVGs in `auto`.
  - Rust backend prefers pre-rendered Rust SVG already in DOM; if missing, may fallback to Mermaid JS (unless explicitly disabled for that operation).
  - GUI post-processing/tuning is applied only in GUI mode.
- PDF preflight path:
  - JS backend runs PDF-mode Mermaid render and then monochrome/grayscale transform.
  - Rust backend swaps `.mermaid` blocks to cached Rust `pdf` SVGs and then applies the same print-safe monochrome/grayscale normalization pass.

Critical invariant

- Never reuse GUI-adjusted Mermaid SVG as Rust PDF output.
- Never select Mermaid cache mode `auto` during Rust PDF export.
- Never leave PDF-mode Rust SVG in place after export completes.
- Restore path (`_restore_preview_mermaid_palette`) must force mode `auto` reapplication. For Rust, this means replacing with cached `auto` SVGs (or regenerating if unavailable).

Practical maintenance guidance

- If you change Mermaid DOM wrappers/toolbars/viewport behavior:
  - Re-test GUI render, PDF export, and post-PDF return for both JS and Rust backends.
- If you change cache keys:
  - Keep mode separation (`auto` vs `pdf`) and hash normalization stable.
- If you change preprint normalization:
  - Ensure toolbars/scrollbars are hidden in PDF.
  - Ensure width/height normalization does not break GUI restore.

Render Triage, Architecture, And Debugging

This section consolidates the former `RENDER-PATHS.md` content into the canonical agent guide.

Use this section before changing Mermaid, PlantUML, preview caching, or PDF export.

Support code has been split into `mdexplore_app/`, but render orchestration still lives primarily in `mdexplore.py`. In practice:

- `mdexplore.py` still owns preview HTML generation, cache injection, and all in-page JavaScript orchestration.
- `mdexplore_app/tree.py` and `mdexplore_app/tabs.py` now own the extracted tree-pane and tab-bar UI support classes, but not render control flow.
- `mdexplore_app/pdf.py` owns PDF footer stamping and related PDF post-pass helpers.
- `mdexplore_app/workers.py` owns the background worker classes used by the window.

When tracing a render bug, start in `mdexplore.py` for control flow and then drop into `mdexplore_app/*` only for leaf helpers.

Preview-only zoom (`Ctrl++`, `Ctrl+-`, `Ctrl+0`) is intentionally outside the diagram render/cache forks described below. It is implemented as a `QWebEngineView.setZoomFactor()` adjustment in `mdexplore.py`, so it scales the entire preview surface after HTML/diagram rendering rather than creating a separate render branch or cache namespace.

Preview overlay markers are also intentionally outside the diagram render/cache forks. Search-hit markers, persistent-highlight markers, and named-view home markers are lightweight in-page overlays driven from already-known match spans or persisted source-line anchors. They should be maintained as post-render UI navigation aids, not as renderer outputs or cache buckets.

0. One-Page Triage Card

Use this as the fast incident-response path before going deep.

Quick Procedure

1. Reproduce on one file with Mermaid diagrams.
2. Run the 4-case matrix in order:
    - JS GUI
    - JS PDF
    - Rust GUI
    - Rust PDF
3. Classify failure scope:
    - only GUI,
    - only PDF,
    - only Rust,
    - only JS,
    - all branches.
4. Jump directly to the matching code waypoints in section `10.3`.
5. Verify fix with both:
    - one small-diagram file,
    - one large multi-diagram file.

Command Checklist (Copy/Paste)

Use these commands from the project root (`/home/npepin/Projects/mdexplore`). For launcher commands rooted at a directory, select the named file in the tree after launch.

```
# 1) Launch with JS backend (baseline)
./mdexplore.sh --mermaid-backend js test
```

```
# 2) Launch with Rust backend (target branch)
./mdexplore.sh --mermaid-backend rust test
```

```
# 3) Force explicit Rust binary if needed
MDEXPLORE_MERMAID_RS_BIN=/home/npepin/.cargo/bin/mmdr \
  ./mdexplore.sh --mermaid-backend rust test
```

```
# 4) Launch rooted at test/ and open committed Mermaid fixture UML.md
./mdexplore.sh --mermaid-backend rust test
```

```
# 5) Launch rooted at repo root and open the larger Mermaid-heavy compatibility stub or
DEVELOPERS-AGENTS.md
./mdexplore.sh --mermaid-backend rust .
```

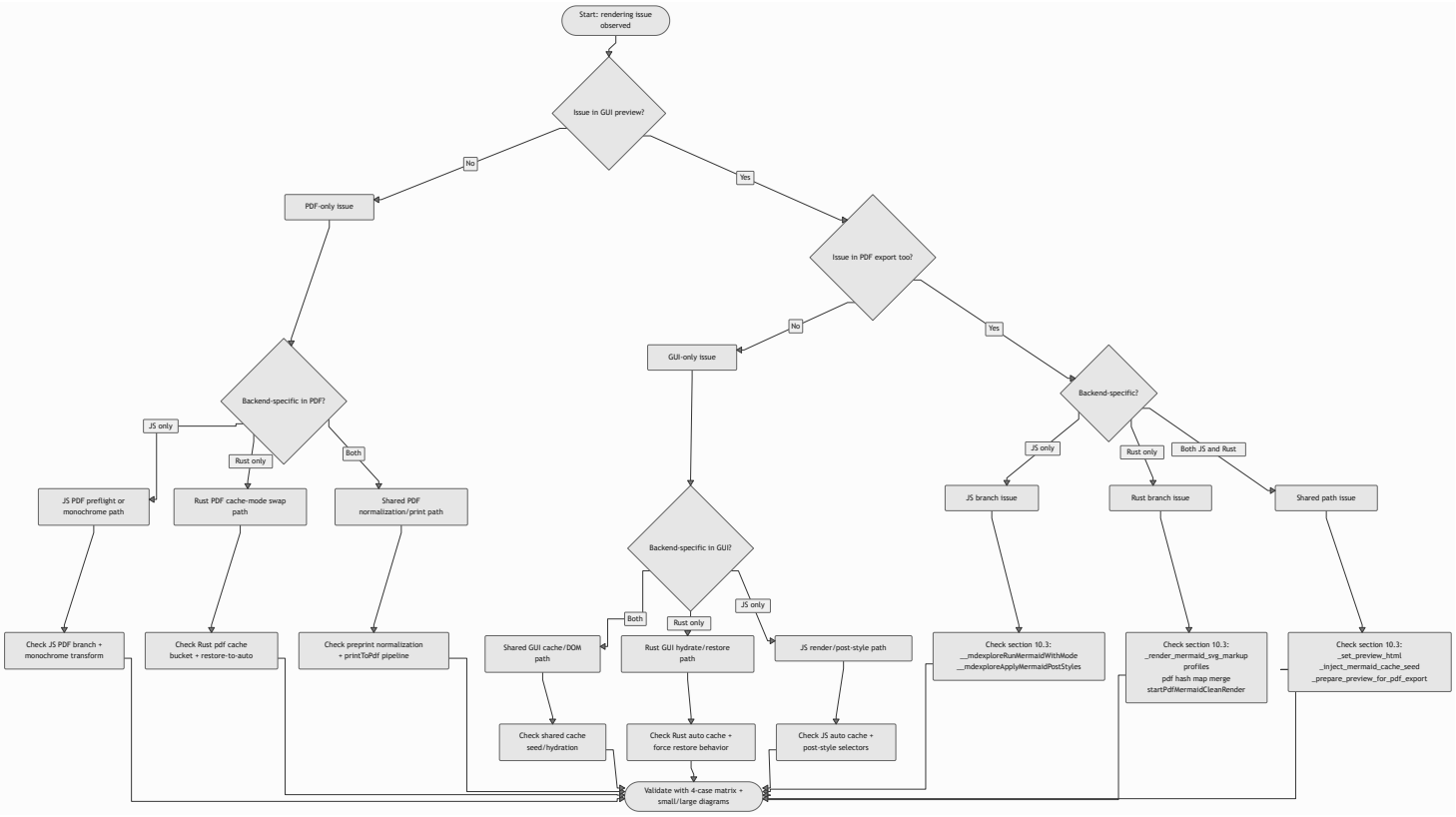
```
# 6) Fast quality gates after code edits
bash -n mdexplore.sh
python3 -m py_compile mdexplore.py
```

Standard Verification Sequence

For each backend (`js` , `rust`) and each file (small + large):

1. Open file and confirm GUI diagram colors/layout.
2. Export PDF and confirm PDF Mermaid path for that backend.
3. Return to GUI and confirm preview styling is restored.
4. Re-open same file in same app run and verify cache path still behaves.

Decision Tree



Immediate Red Flags

- Rust PDF output looks like GUI output.
  - likely mode mix-up (`auto` used where `pdf` expected).
- GUI remains white/print-styled after PDF export.
  - likely restore-to- `auto` did not reapply.
- Mermaid in PDF is tiny relative to page width.
  - likely width/height attr normalization incomplete.
- Only large diagrams fail.
  - likely `setHtml` payload/temporary-file load path and timing interactions.

1. Scope and Intent

`mdexplore` has multiple render branches:

- GUI preview vs PDF export mode.
- Mermaid JS backend vs Rust Mermaid backend (`mmdr`).
- Python-side render/cache responsibilities vs in-page JavaScript responsibilities.

The main maintenance risk is crossing these concerns and accidentally reusing the wrong SVG variant (for example reusing GUI-adjusted SVGs in PDF mode).

This section does not treat the following as render branches:

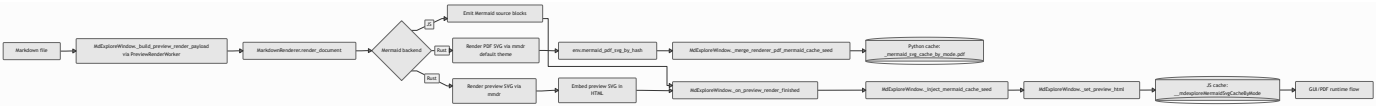
- preview-wide `QWebEngineView` zoom,
- preview overlay markers for search hits / persistent highlights / named views,
- tree gutter badge composition (search pill, highlight marker, views badge).

Those features consume render outputs, but they do not create alternate HTML, diagram SVG, or cache namespaces.

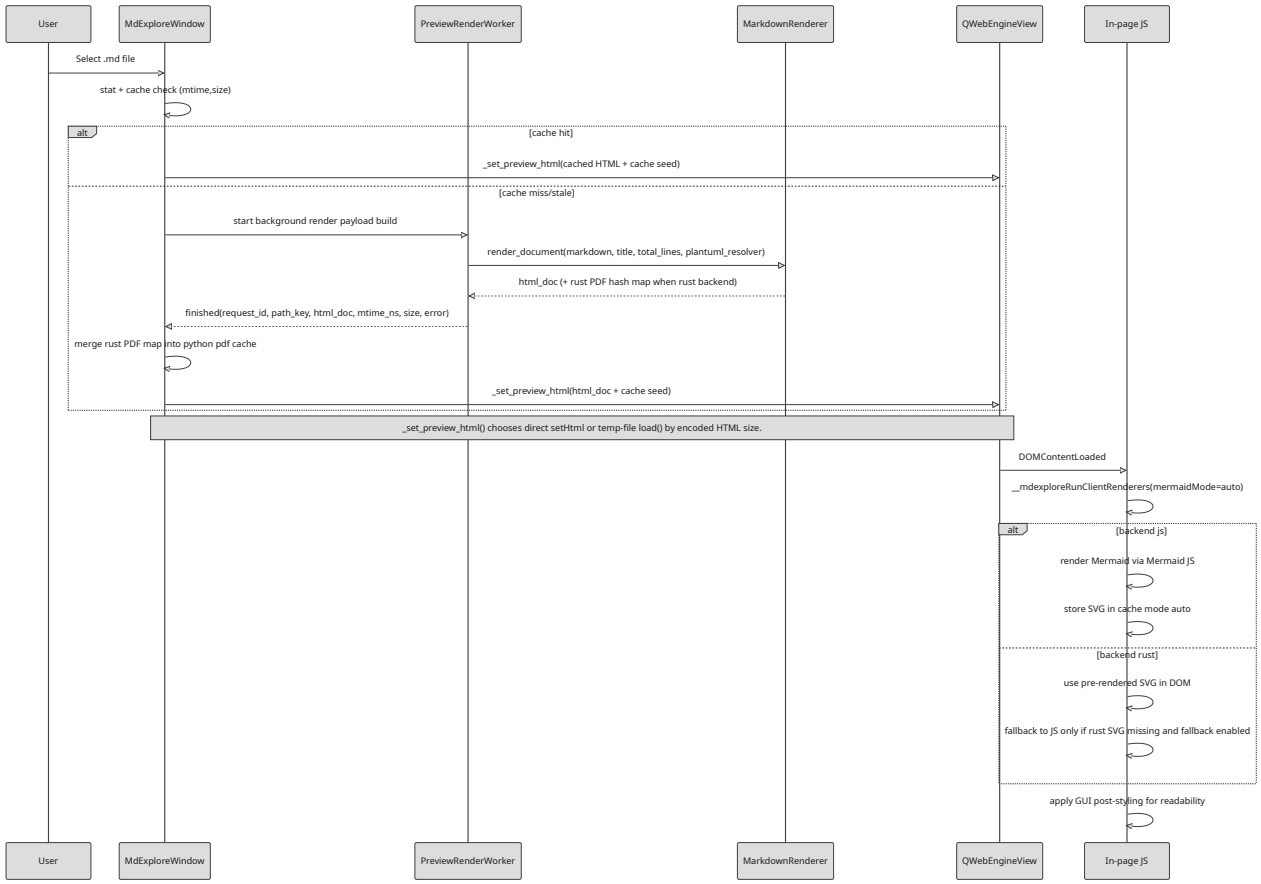
2. Render Mode Matrix

Mode	Mermaid Backend	SVG Source	Post-processing	Primary Cache Bucket
GUI	JS	Mermaid JS runtime (`mermaid.render`)	GUI contrast/style tuning	`auto`
GUI	Rust	`mmdr` preview profile (Python)	GUI contrast/style tuning	`auto`
PDF	JS	Mermaid JS runtime in PDF preflight	print monochrome/grayscale transform	`pdf`
PDF	Rust	`mmdr` PDF profile (default theme, Python)	print-safe grayscale normalization (multi-shade)	`pdf`

3. End-to-End Ownership



4. GUI Preview Path



4.1 Preview Overlay Layers (Non-Render)

After the HTML is already loaded, mdexplore may add three navigation overlays inside the preview viewport:

- right-side yellow search-hit markers derived from the active search result DOM,
- left-side persistent-highlight markers derived from persisted text highlight spans, with separate normal and important visual styles,
- left-side named-view markers derived from saved tab home line numbers.

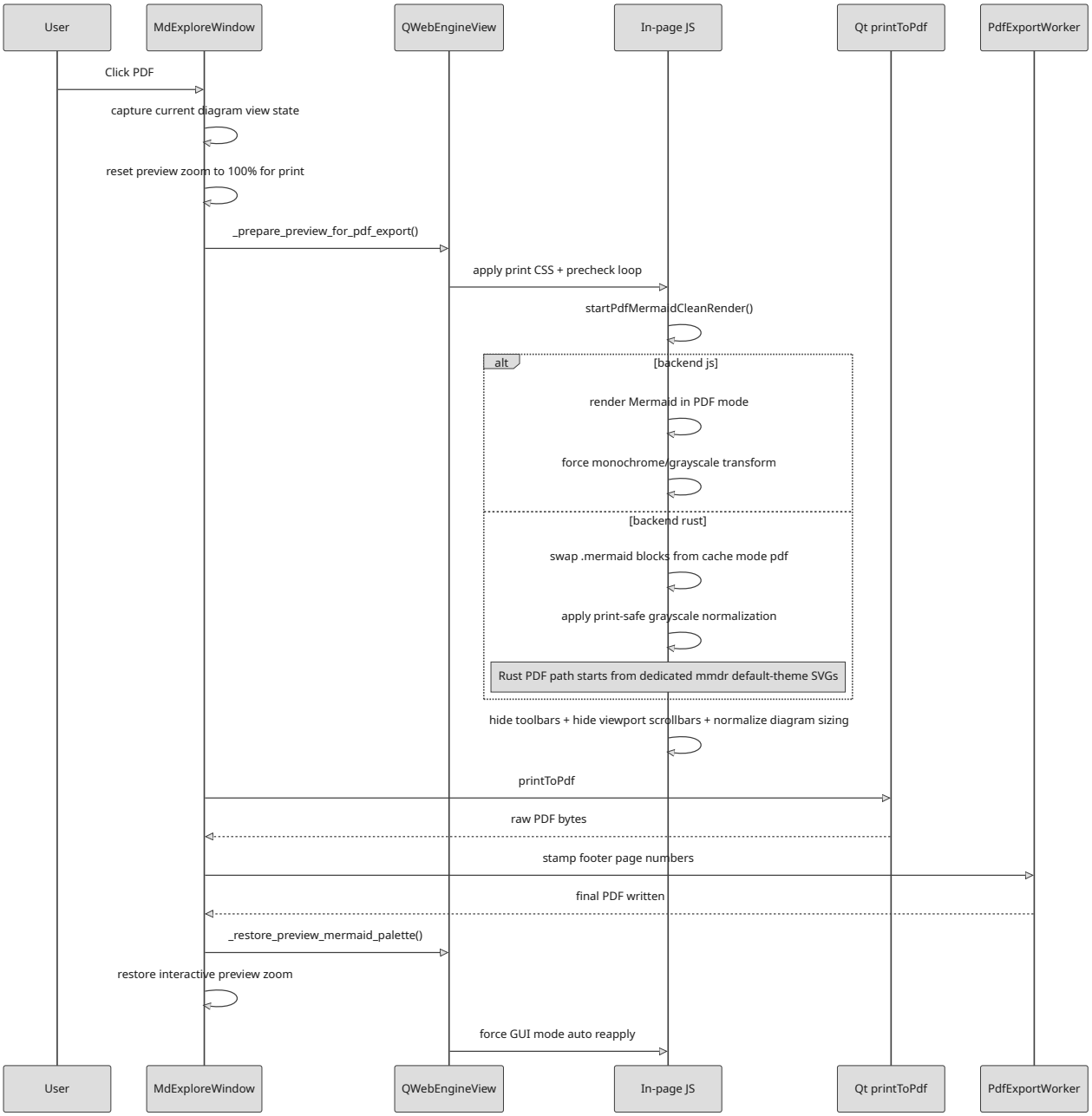
These overlays are intentionally kept outside the diagram render branches:

- search markers are rebuilt from existing highlighted-match DOM,
- persistent-highlight markers are rebuilt only when persisted highlight spans exist,
- named-view markers are pushed from Python as a compact line-anchor payload and do not scan the DOM.

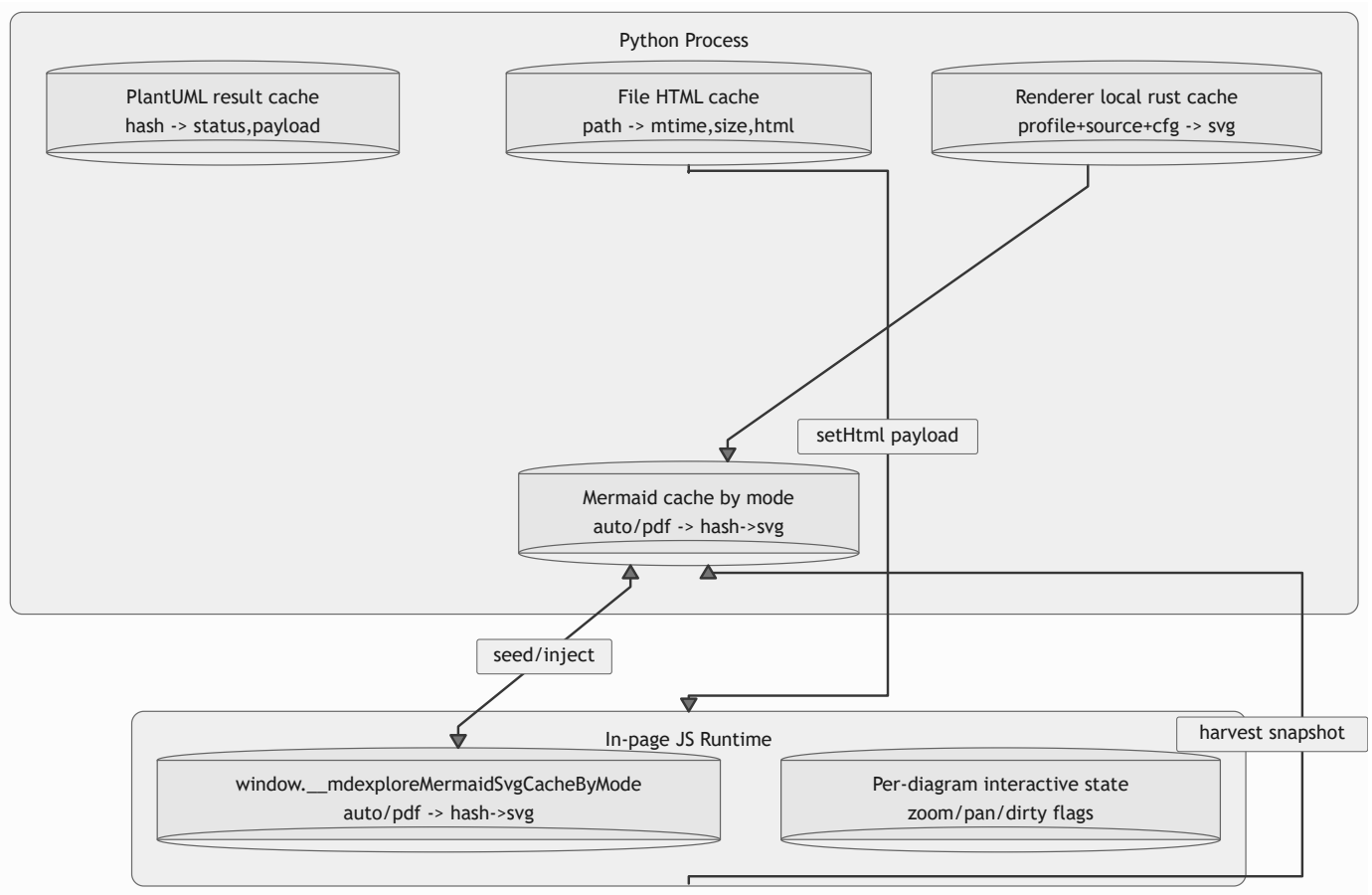
The layering order is significant:

1. persistent-highlight markers,
2. named-view markers on top when positions overlap.

5. PDF Export Path



6. Cache Architecture



Oversized preview HTML uses `_set_preview_html()` temp-file transport instead of direct `setHtml()`, but that is only a load path and not a separate cache bucket.

7. Critical Invariants

- Keep Mermaid cache separation by mode (`auto` vs `pdf`).
- Rust PDF output must come from the dedicated Rust PDF render profile, not GUI SVG.
- GUI Mermaid post-processing must not run on Rust PDF SVGs.
- For PDF pagination, headed Mermaid and PlantUML sections are classified on the diagram fence itself rather than on an outer wrapper. This avoids the Chromium behavior where a heading can be left behind on the prior page while the diagram starts on the next page.
- Rust PDF path may apply print-safe grayscale normalization after selecting the dedicated Rust PDF SVG source.
- PDF preflight must hide interactive controls (toolbars/scrollbars) before snapshot.
- Post-export restore must force a GUI-mode reapply.

8. Failure and Fallback Rules

- Rust backend unavailable:
  - app falls back to JS backend and reports warning.
- Rust render missing SVG for a block:
  - GUI path may JS-fallback unless disabled for that operation.
  - PDF path reports explicit Mermaid failure for that block if no PDF SVG exists.
- PlantUML failure:
  - render placeholder becomes detailed error text; markdown stays visible.
- Stale cache safety:
  - file cache keyed by `mtime_ns` and `size`; changed files are re-rendered.

9. Known TODO

- Diagram zoom/pan state restore across document switches is still not fully reliable for all Mermaid/PlantUML navigation sequences in one app run.
- See `DEVELOPERS-AGENTS.md` for attempted approaches and constraints.
- Preview-only zoom factor is intentionally session-local and is not persisted per file, per view, or into any render cache bucket.

10. Debugging Playbook (Human + Agent)

This section is a practical troubleshooting guide for render regressions. Use it when behavior deviates from expectations.

10.1 First Principles

- Always isolate by branch first:
  - GUI vs PDF.
  - Mermaid backend `js` vs `rust`.
  - Small diagram vs large diagram.
- Keep one variable changing per test run.
- Verify whether failure is:
  - source generation,
  - cache selection,
  - in-page transform,
  - restore after PDF.

10.2 Fast Isolation Matrix

Run these four cases on the same markdown file with Mermaid:

Case	Backend	Path	Expected
A	JS	GUI	dark-theme tuned Mermaid in preview
B	JS	PDF	monochrome/grayscale Mermaid in PDF
C	Rust	GUI	dark-theme tuned Mermaid in preview
D	Rust	PDF	dedicated Rust PDF SVG + print-safe grayscale normalization

If only one case fails, debug that branch first.
If all fail, suspect shared setup (asset loading, cache seed injection, or broken JS runtime).

10.3 Key Code Waypoints

Use these anchors in `mdexplore.py` while debugging:

- Mermaid render profile split:
  - `MarkdownRenderer._render_mermaid_svg_markup(..., render_profile=...)`
- Rust PDF SVG capture:
  - `env["mermaid_pdf_svg_by_hash"]` in `render_document()`
  - `take_last_mermaid_pdf_svg_by_hash()`
  - `MdExploreWindow._merge_renderer_pdf_mermaid_cache_seed()`
- Background preview render handoff:
  - `_build_preview_render_payload(...)`
  - `_on_preview_render_finished(...)`
- HTML injection path:
  - `_inject_mermaid_cache_seed(...)`
  - `_set_preview_html(...)`
- In-page runtime orchestration:
  - `window.__mdexploreRunMermaidWithMode(...)`
  - `startPdfMermaidCleanRender(...)`
  - `window.__mdexploreApplyMermaidPostStyles(...)`
- PDF mode entry/exit:
  - `_prepare_preview_for_pdf_export(...)`
  - `_trigger_pdf_print(...)`
  - `_restore_preview_mermaid_palette(...)`

10.4 Symptom -> Likely Cause -> Next Check

Symptom	Likely Cause	Next Check
Rust PDF still shows GUI colors	wrong SVG source reused	verify Rust PDF path is pulling cache mode `pdf`, not current DOM
GUI stays white after PDF (Rust)	restore did not reapply `auto` SVG	verify `_restore_preview_mermaid_palette` forces mode `auto` and Rust branch honors `force`
PDF Mermaid tiny on page	width/height attrs not normalized	verify print preflight strips width/height attrs and sets width/max-width/height
GUI Mermaid labels too dark	post-style function missed selector class	inspect `__mdexploreApplyMermaidPostStyles` for diagram-kind-specific logic
Random "Preview load failed" on heavy files	`setHtml` payload too large / transient web load issue	verify temp-file fallback path in `_set_preview_html` and check status/log output
Rust fallback unexpectedly using JS in PDF	fallback guard missing in PDF stage	verify Rust PDF branch does not trigger JS fallback render path
Sequence diagram labels invisible in PDF	label content not normalized to dark text	verify grayscale pass handles both SVG text and foreignObject HTML labels

10.5 Runtime Inspection Steps (Inside Running App)

When the preview is open, inspect these JS objects from the page context:

- `window.__mdexploreMermaidBackend`
- `window.__mdexploreMermaidSvgCacheByMode`
- `window.__mdexploreMermaidPaletteMode`
- `window.__mdexplorePdfMermaidReady`
- `window.__mdexplorePdfMermaidError`

Pedagogical check sequence:

1. Confirm backend matches intent (`js` or `rust`).
2. Confirm both cache buckets exist (`auto` and `pdf`).
3. Confirm target diagram hash exists in expected bucket.
4. Confirm SVG payload in that bucket actually contains `<svg`.
5. Confirm active Mermaid mode (`auto` / `pdf`) matches current operation.

10.6 Cache and Ownership Sanity Rules

Treat these as hard assertions:

- Python cache ownership:
  - source of truth for injected seed snapshot at page load.
- JS cache ownership:
  - source of truth for in-page re-render/re-hydrate operations.

- Cross-mode safety:
  - `auto` and `pdf` caches must never be mixed.
- Rust profile safety:
  - preview profile output and PDF profile output are intentionally different artifacts.

10.7 Practical Reproduction Protocol

Use this exact sequence to reproduce and verify fixes:

1. Launch with JS backend and test one file (`GUI -> PDF -> GUI`).
2. Launch with Rust backend and repeat same file (`GUI -> PDF -> GUI`).
3. Repeat with:

- one small Mermaid diagram file,
- one large multi-diagram file.

4. For Rust, verify:

- GUI returns to dark-tuned view after PDF export,
- PDF starts from dedicated Rust PDF SVGs and remains readable after grayscale normalization.

5. Re-open the file in same app session and ensure cache path still behaves.

10.8 Logging and Instrumentation Strategy

When uncertain, add temporary, narrow logging at branch points:

- entering/exiting PDF mode,
- selecting Mermaid mode (`auto` / `pdf`),
- backend switch (`js` / `rust`),
- hash lookup misses per mode,
- restore completion path.

Remove or gate verbose logs after confirming the fix.

10.9 Acceptance Criteria for Render Fixes

A Mermaid/PDF fix is complete only when all are true:

- No regression in JS GUI preview.
- No regression in Rust GUI preview.
- JS PDF still produces readable print-safe output.
- Rust PDF uses dedicated Rust PDF SVG path before grayscale normalization.
- GUI view restores correctly after PDF export.
- Behavior holds for both small and large diagrams.
- Documentation (`README.md`, `DEVELOPERS-AGENTS.md`) reflects the final path.

Known TODO: Diagram View State Restore

- Open issue: Mermaid and PlantUML zoom/pan state is not yet reliably restored when switching away from a document and then returning during the same app run.
- Treat this as a known TODO. Do not remove this note until behavior is verified end-to-end and documented with a reproducible test.

Attempts Tried That Did Not Fully Resolve It

- Frequent JS-side state capture timers plus explicit capture on file-switch.
- Per-diagram state keys and in-page reapply hooks (`__mdexploreReapplySavedState`).
- Multi-pass deferred restore reapply from Python (`QTimer` at staggered delays).
- Reapplying restore after Mermaid palette reset and after async renderer completion.
- Forcing PDF-mode normalization/reset before print and restoring interaction state afterward.

These approaches improved behavior in some cases but did not make restore reliable across all navigation sequences.

Launcher Rules

- `mdexplore.sh` must be runnable from any working directory.
- It must keep venv handling deterministic:
  - create if missing
  - install dependencies
  - run app
  - invoke `.venv/bin/python` directly (no shell activation required)
  - do not use `exec -a` with Python (can break venv resolution)
- Launcher arg handling must tolerate `.desktop` `%u` behavior:
  - empty argument should behave like "no path"
  - `file://` URIs should be decoded
  - file arguments should resolve to parent directory
- Launcher must pass through supported app switches (for example `--mermaid-backend`).
- Launcher `DEBUG_MODE=true` should append `--debug` so application debug logging can be enabled without editing desktop launchers or shell aliases.
- Non-interactive launcher runs should log to `~/.cache/mdexplore/launcher.log` for desktop troubleshooting. Log retention is capped to the most recent 1000 lines.
- Launcher should verify key runtime imports (`markdown_it` , `linkify_it` , `PySide6.QtWebEngineWidgets` , `pypdf` , `reportlab.pdfgen.canvas`) and self-heal by reinstalling requirements if the environment is incomplete.
- `--help` should stay lightweight and not require venv activation.

Setup Script Rules

- `setup-mdexplore.sh` is the full local bootstrap path and should remain safe to rerun.
- It should:
  - create/update `.venv`
  - install Python dependencies from `requirements.txt`
  - ensure vendored local assets exist for MathJax, Mermaid JS, and PlantUML
  - ensure the Rust Mermaid renderer source exists under `vendor/mermaid-rs-renderer`
  - build `mmdr` into the path already probed by the launcher/app
- If the vendored Rust renderer source already exists, prefer it over recloning.
- If `cargo` is missing, the setup script may install a minimal Rust toolchain through `rustup` so bootstrap can complete non-interactively.
- The setup script should not silently apt-install arbitrary system packages. Missing optional system runtime pieces (for example Java or `code`) should produce clear warnings instead.

Quality Gates Before Finishing

Run at least:

```
bash -n mdexplore.sh
python3 -m py_compile mdexplore.py
```

If behavior changes, also run manual smoke tests:

1. Launch with no path.
2. Launch with explicit path.
3. Open `.md` file and verify render.
4. Verify `Edit` behavior with and without `code` in `PATH`.

Documentation Requirements

When changing behavior:

- Update `README.md` usage and examples.
- Update `DEVELOPERS-AGENTS.md` when behavioral hierarchy, precedence, invariants, render/caching/PDF/diagram control flow, or agent-facing workflow changes.
- Keep `RULES.md` and `RENDER-PATHS.md` compatibility stubs aligned if the canonical section names in `DEVELOPERS-AGENTS.md` move.

Common Feature Patterns

- New keyboard action: add `QAction` in `_add_shortcuts`.
- New toolbar action: add button in top bar setup and handler method.
- New markdown extension: extend custom fence handling in `MarkdownRenderer`.
- New CLI option: update argparse setup in `main`, then mirror wrapper behavior in `mdexplore.sh`.

Out of Scope Unless Requested

- Packaging into Snap/Deb/AppImage.
- Multi-tab editing or embedded text editor.
- Background file watching/auto-reload loops.
- Cross-platform polish outside Linux desktop behavior.